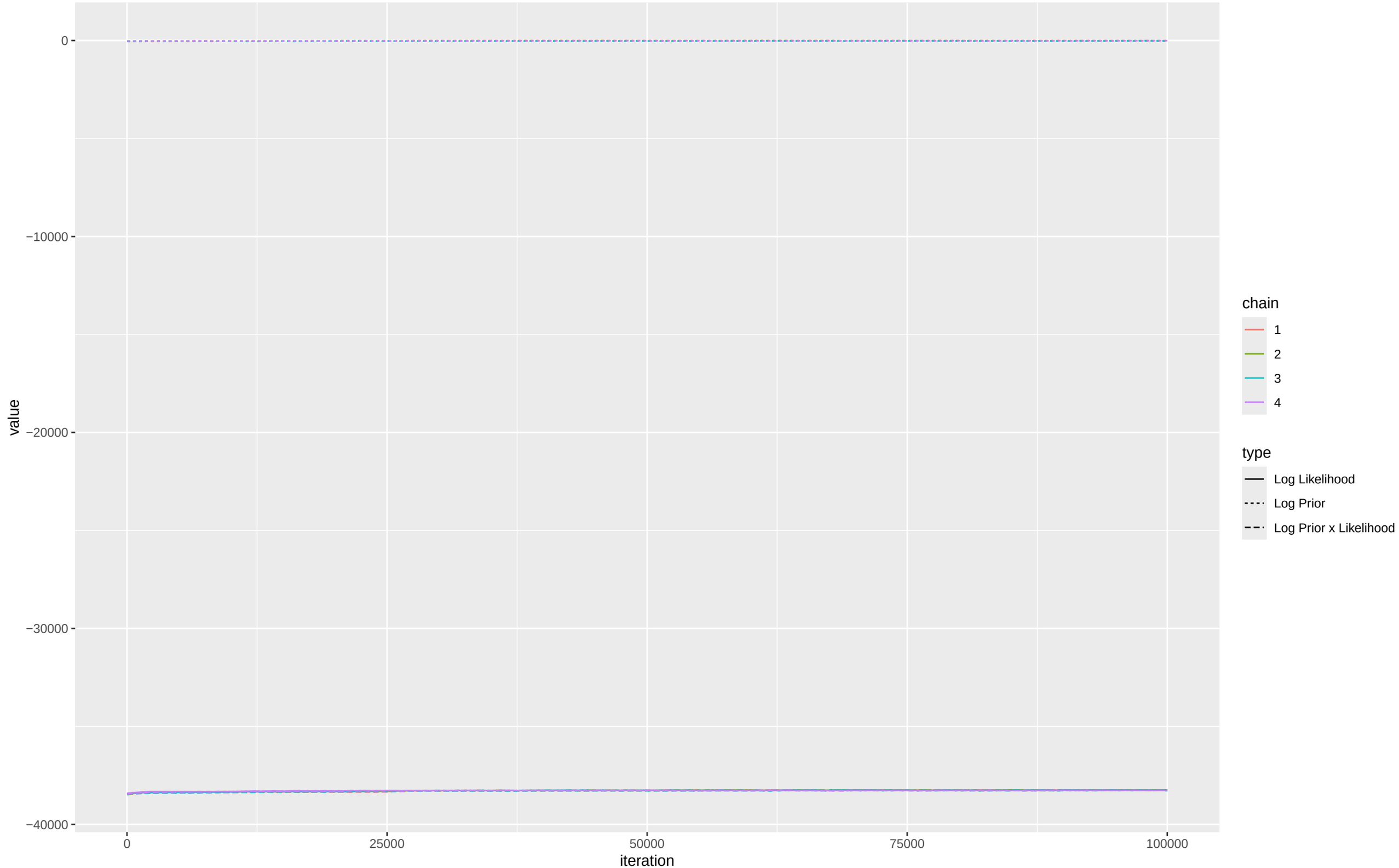
```
r1.lower.middle, step 2 (2026-10-20)

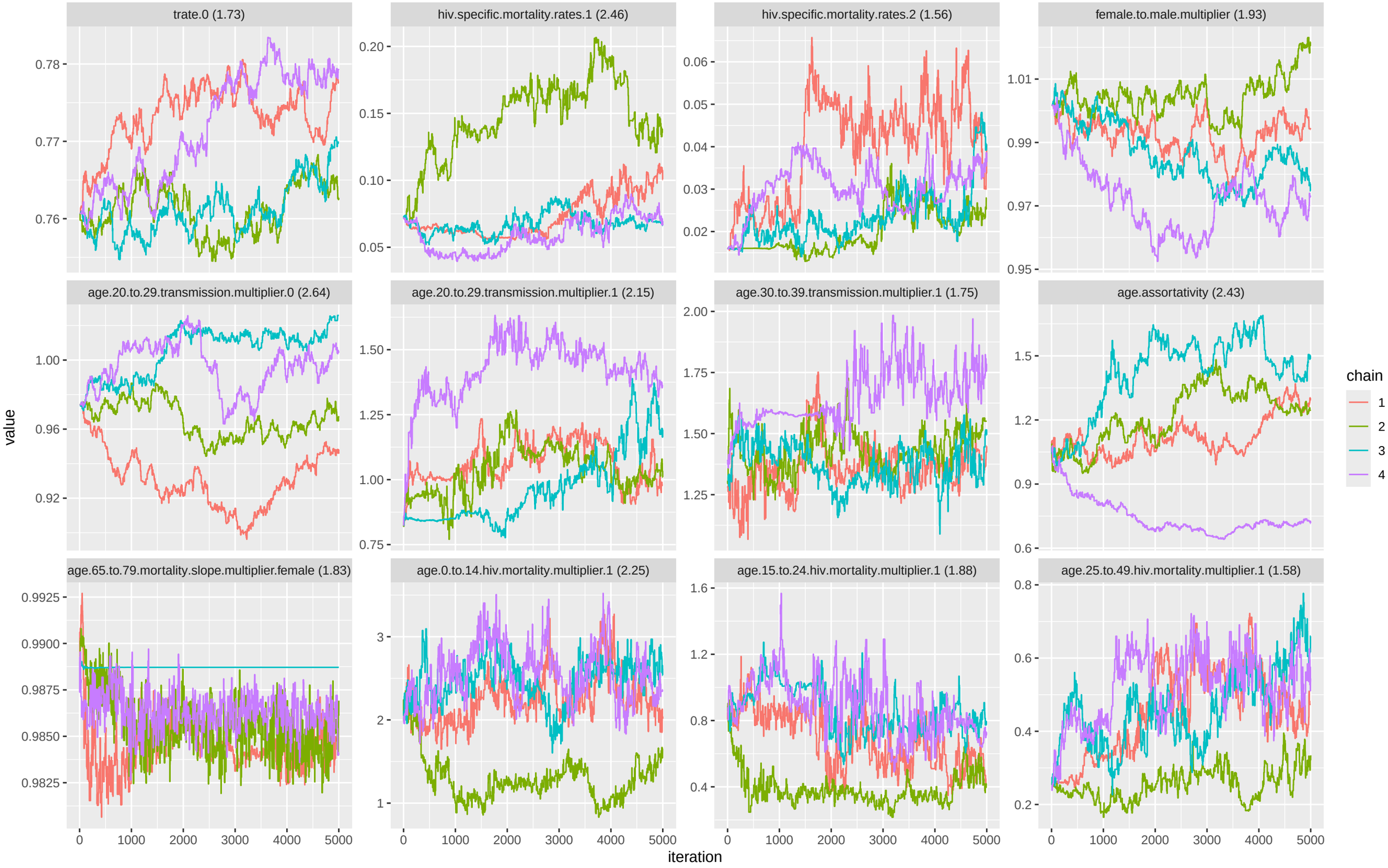
 chain     seed iterations.saved acceptance.rate final.log.likelihood
     1 20263016             5000           0.277              -38260.7
     2 20263017             5000           0.277              -38247.0
     3 20263018             5000           0.276              -38260.1
     4 20263019             5000           0.275              -38257.8
 best.log.likelihood errored.proposals
            -38241.4                 0
            -38238.1                 0
            -38244.5                 0
            -38248.5                 0
                                              file
 mcmc_r1.lower.middle_step2_chain1_2026-10-19.Rdata
 mcmc_r1.lower.middle_step2_chain2_2026-10-20.Rdata
 mcmc_r1.lower.middle_step2_chain3_2026-10-20.Rdata
 mcmc_r1.lower.middle_step2_chain4_2026-10-20.Rdata

Worst R-hats (> 1.1 suggests the chains haven't converged):
            age.20.to.29.transmission.multiplier.0
                                             2.644
                   hiv.specific.mortality.rates.1
                                             2.456
                                 age.assortativity
                                             2.434
            age.0.to.14.hiv.mortality.multiplier.1
                                             2.250
            age.20.to.29.transmission.multiplier.1
                                             2.153
                        female.to.male.multiplier
                                             1.928
       age.15.to.24.hiv.mortality.multiplier.1
                                             1.884
age.65.to.79.mortality.slope.multiplier.female
                                             1.830
        age.30.to.39.transmission.multiplier.1
                                             1.748
                                           trate.0
                                             1.729
       age.25.to.49.hiv.mortality.multiplier.1
                                             1.584
                hiv.specific.mortality.rates.2
                                             1.560
        age.40.to.49.transmission.multiplier.2
                                             1.508
        age.30.to.39.transmission.multiplier.0
                                             1.466
                                           trate.1
                                             1.456
        age.20.to.29.transmission.multiplier.2
                                             1.416
                     age.20.to.24.aging.factor
                                             1.415
        age.40.to.49.transmission.multiplier.0
                                             1.374
                  log.OR.testing.intercept
```

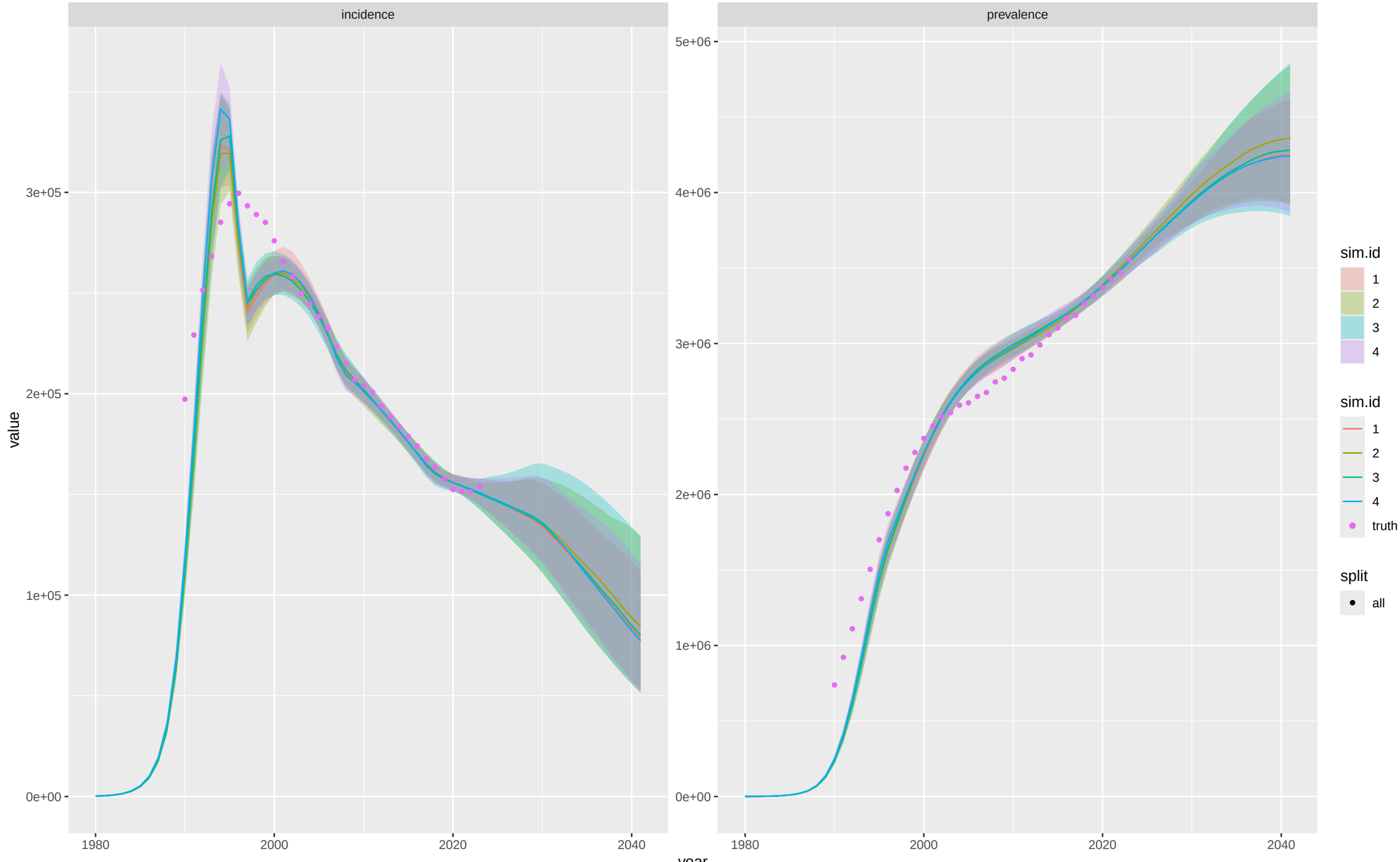
r1.lower.middle step 2: likelihood



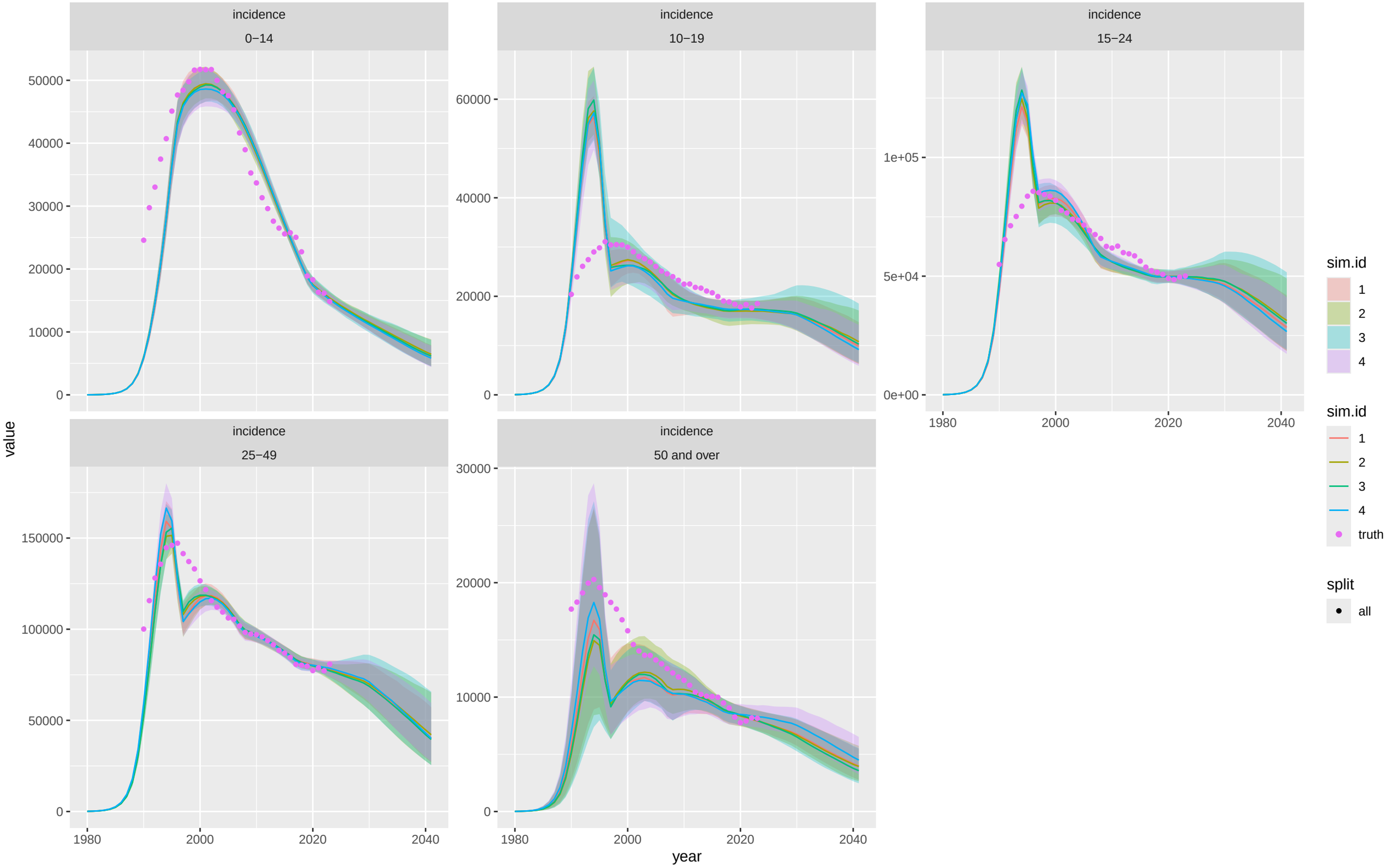
r1.lower.middle step 2: trace



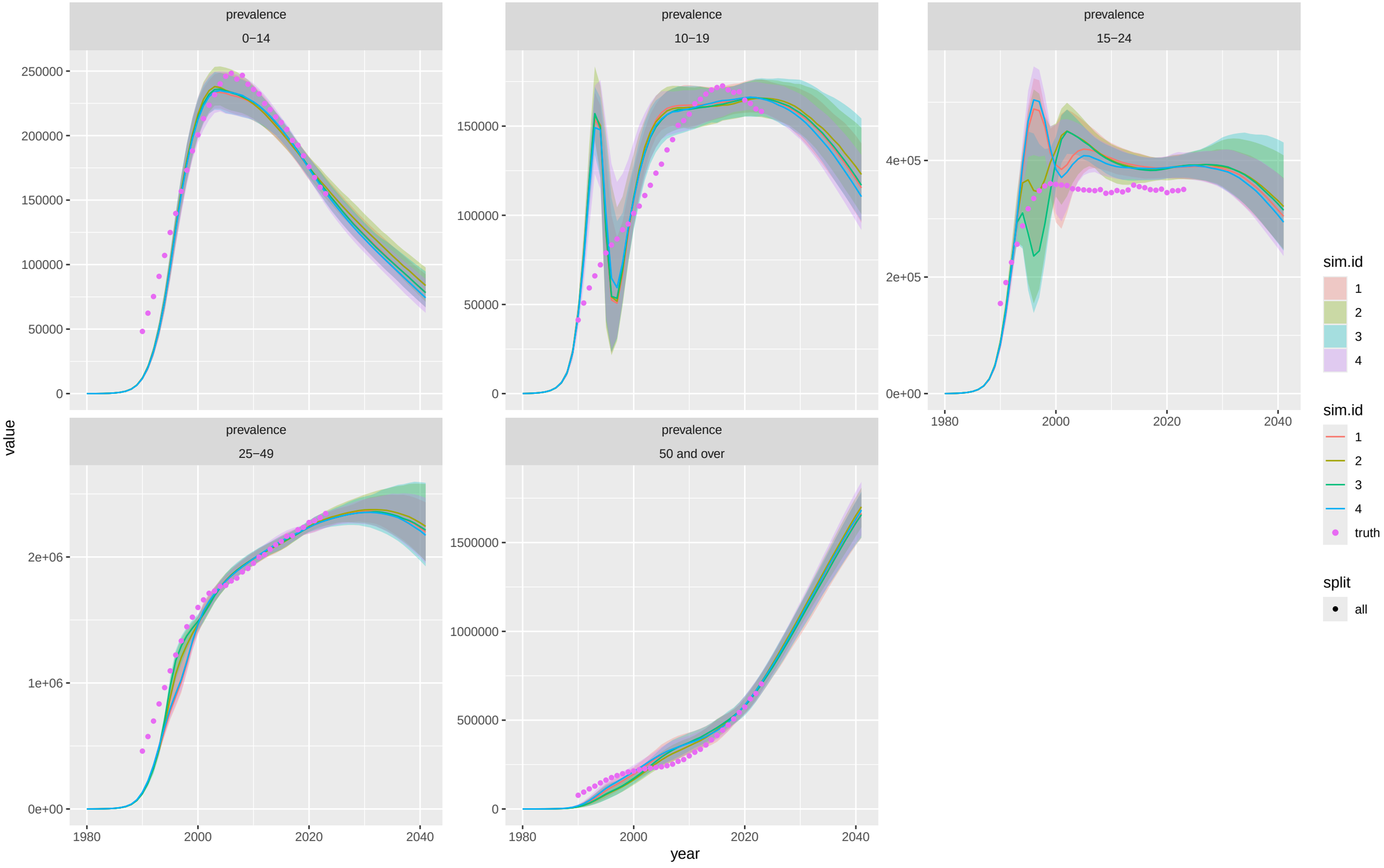
r1.lower.middle step 2: fit (one color per chain)



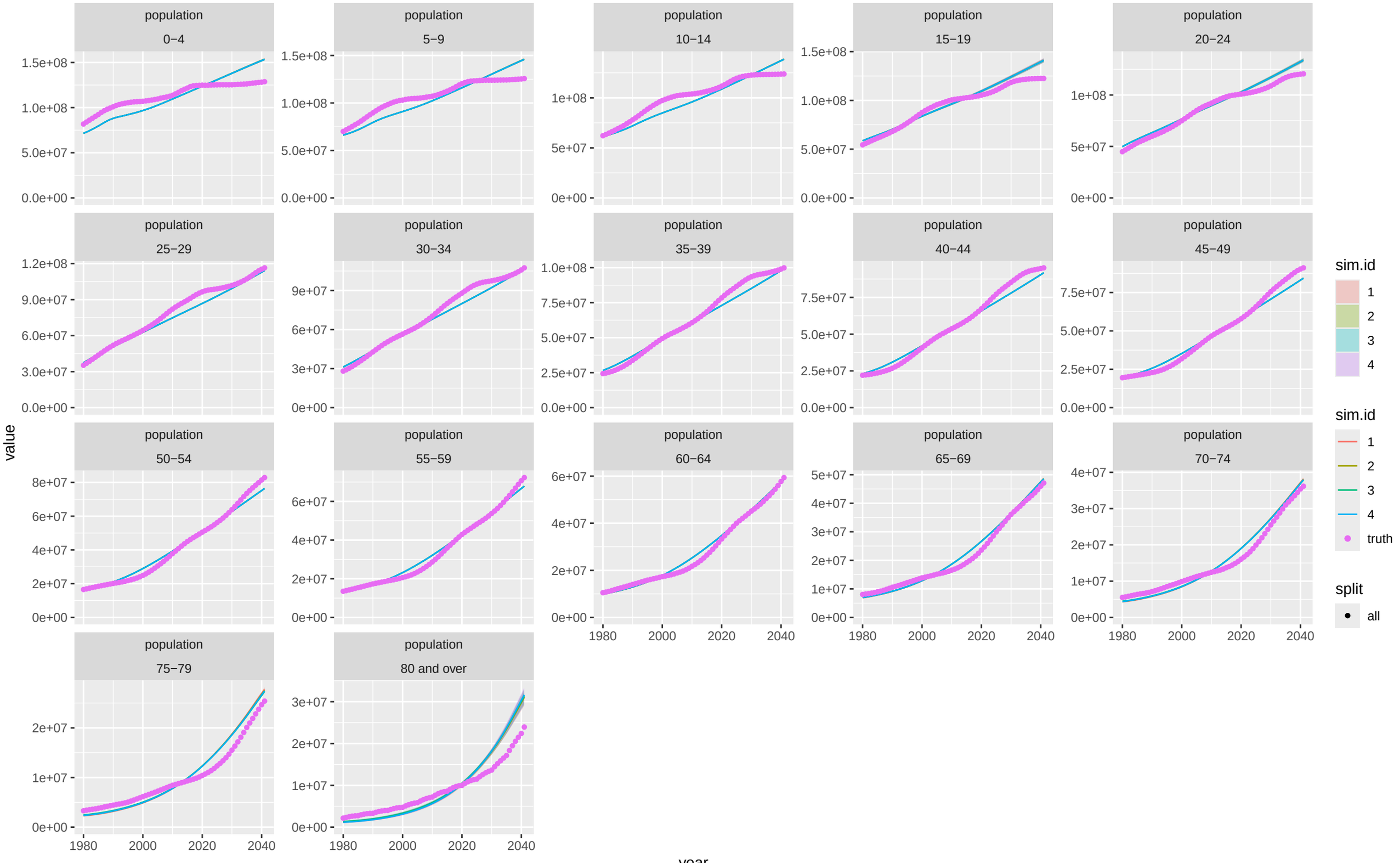
r1.lower.middle step 2: incidence.by.age (one color per chain)



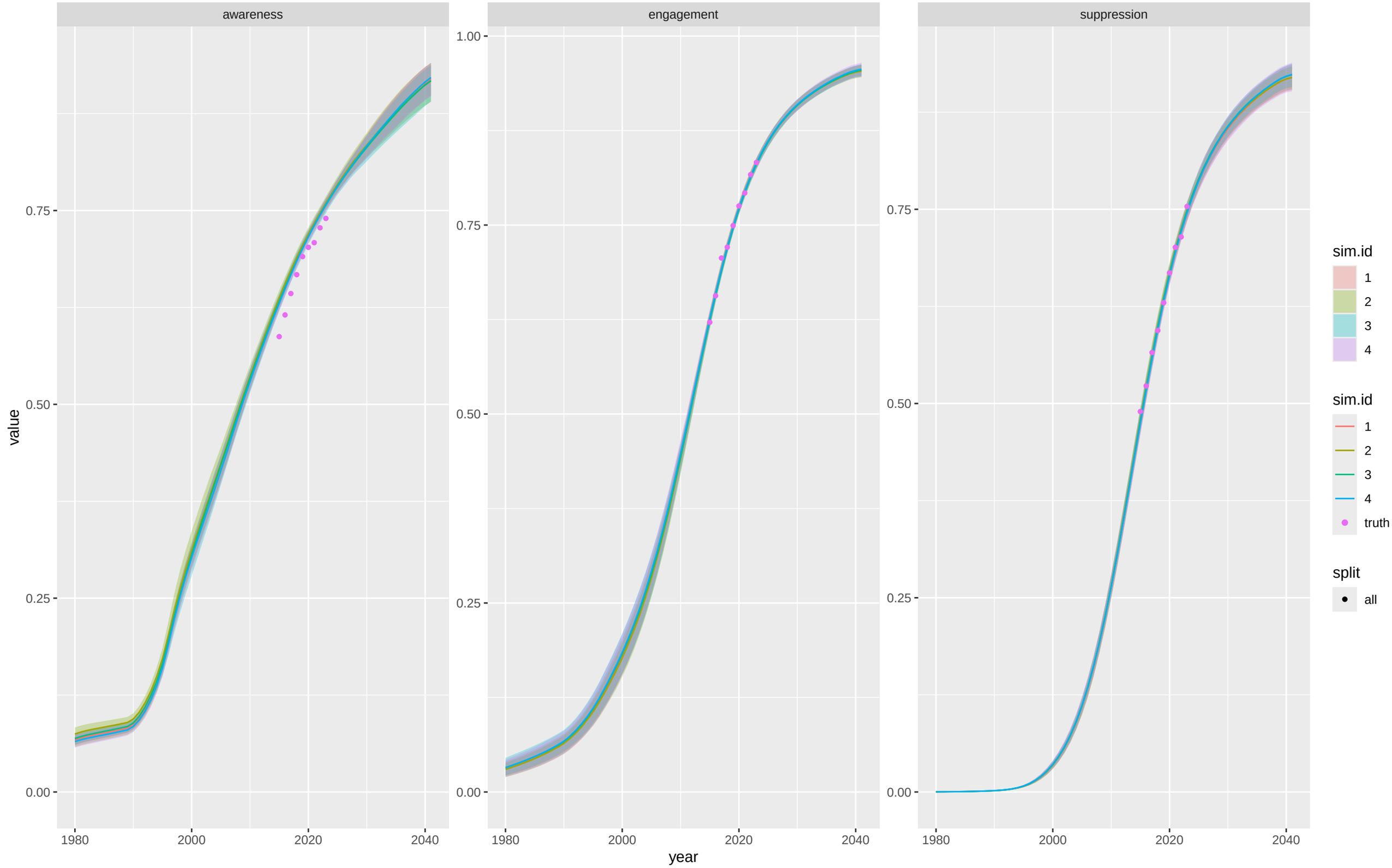
r1.lower.middle step 2: prevalence.by.age (one color per chain)



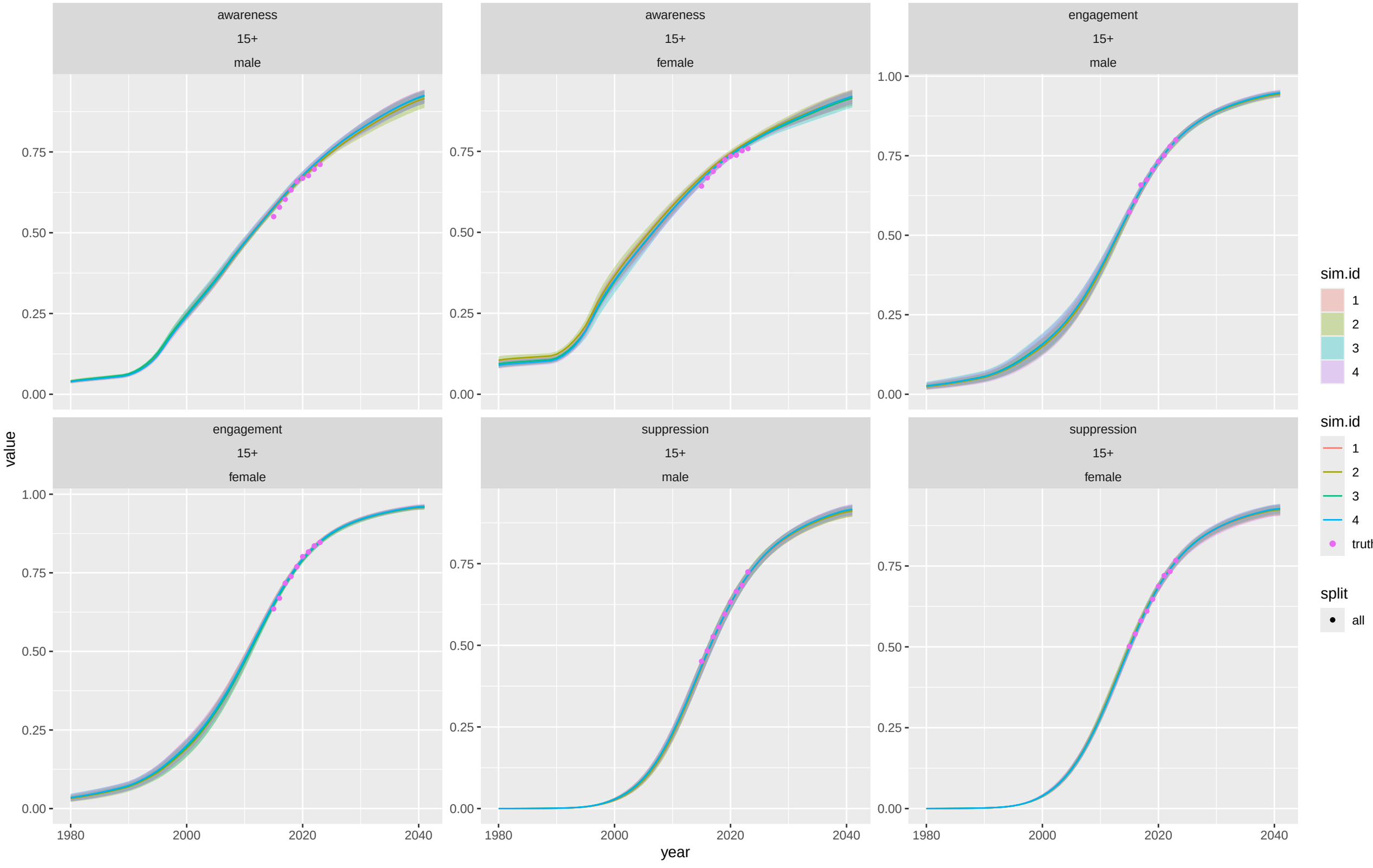
r1.lower.middle step 2: population.by.age (one color per chain)



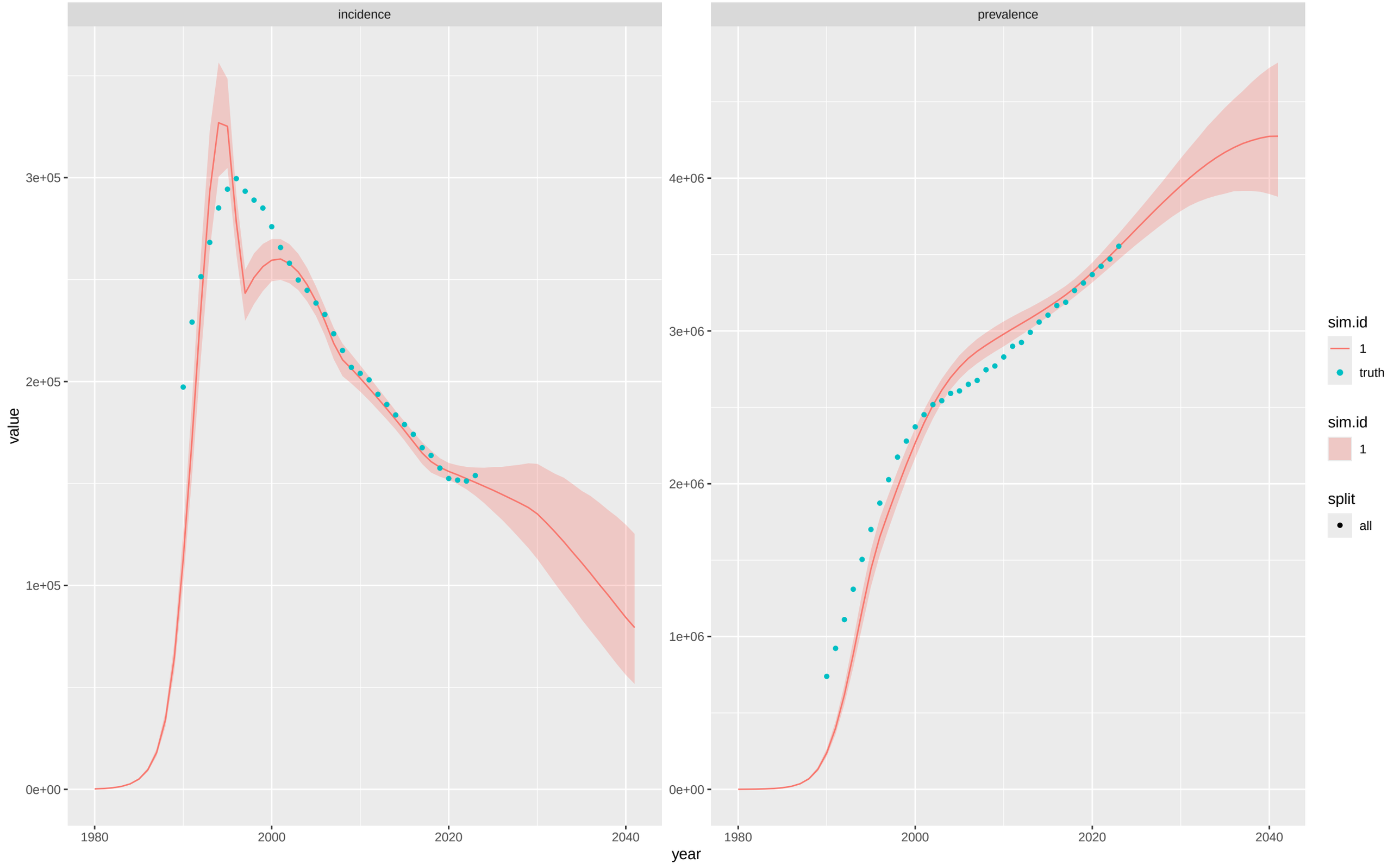
r1.lower.middle step 2: cascade (one color per chain)



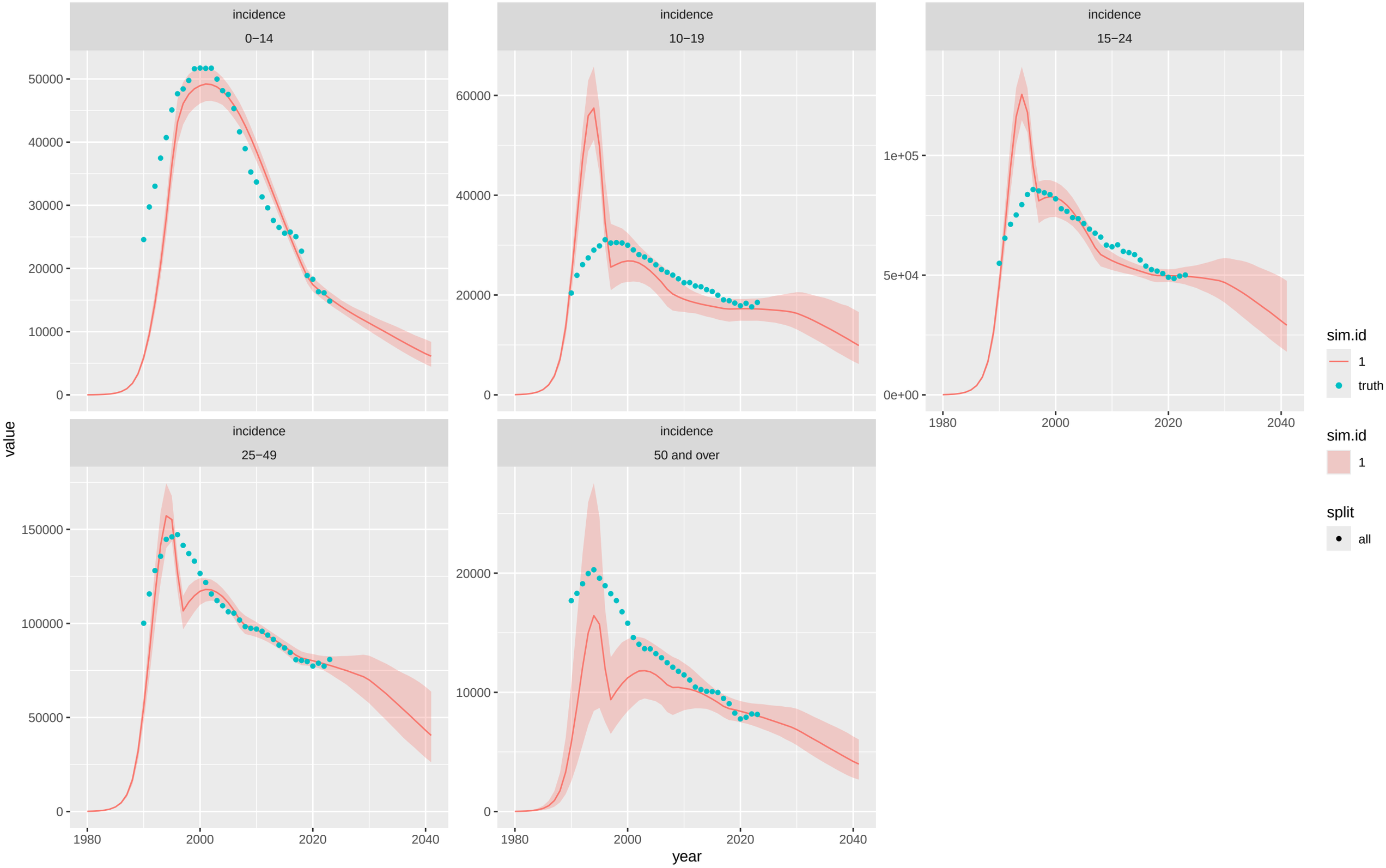
r1.lower.middle step 2: cascade.by.age.sex (one color per chain)



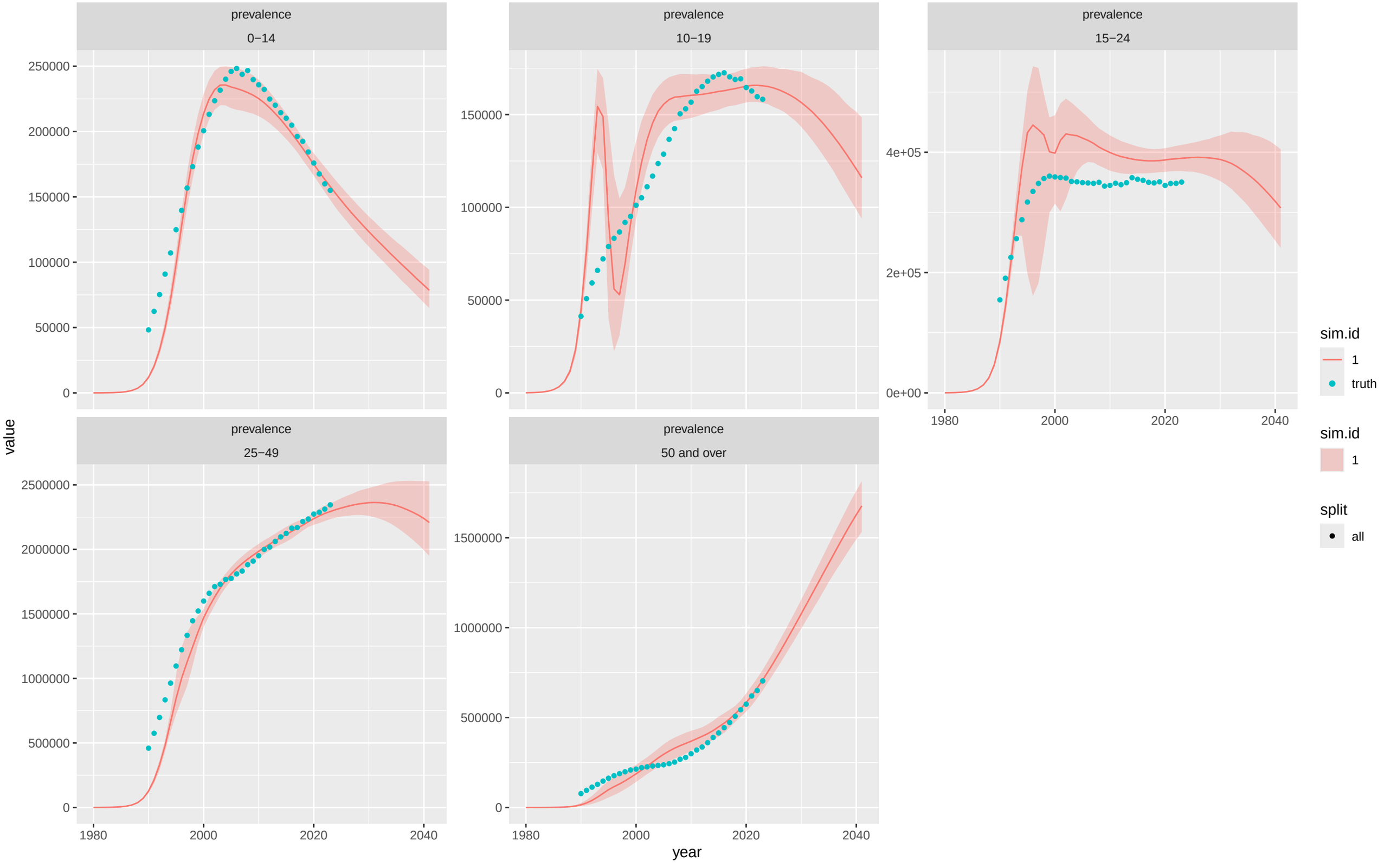
r1.lower.middle: fit (merged Step 2 simset)



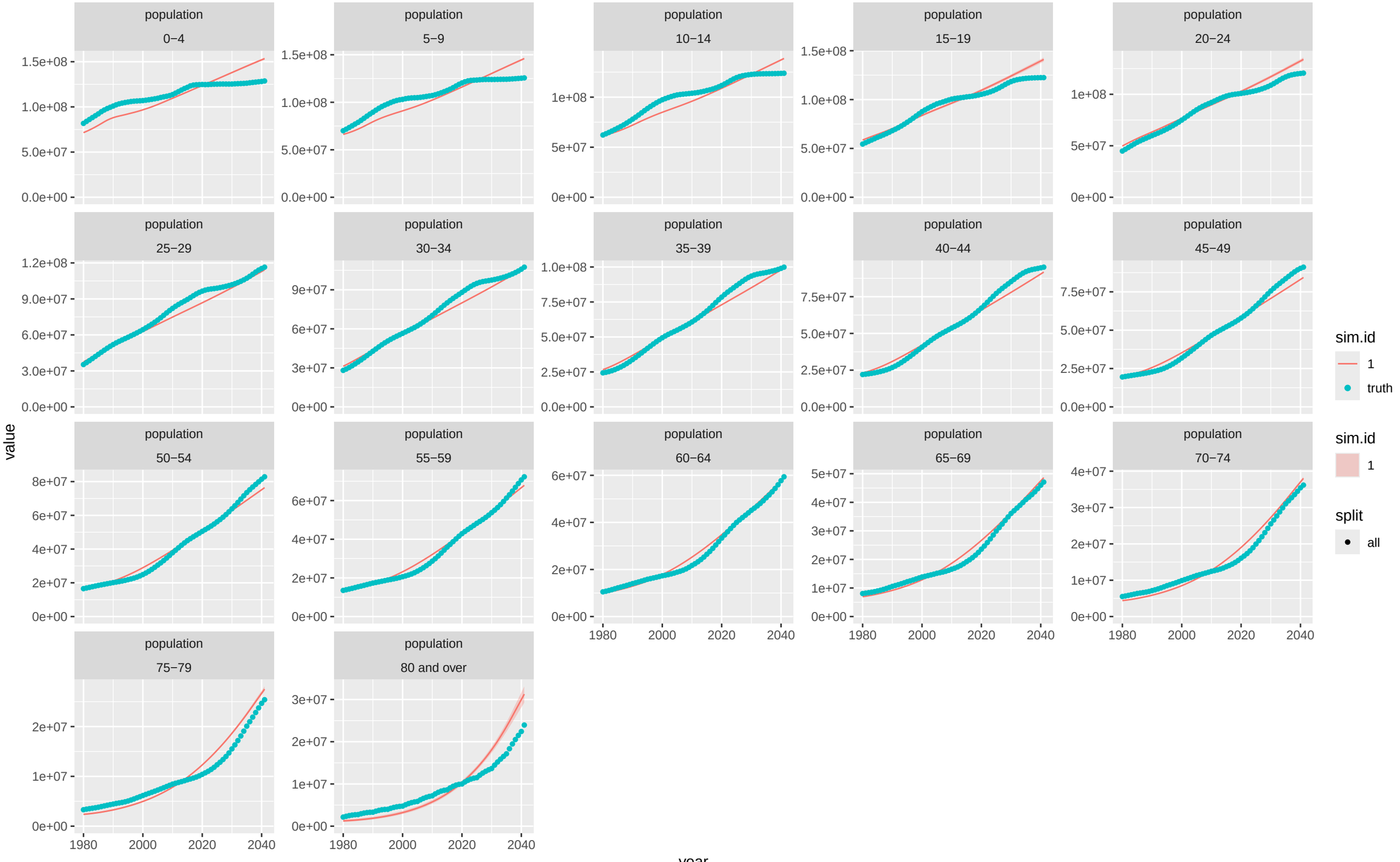
r1.lower.middle: incidence.by.age (merged Step 2 simset)



r1.lower.middle: prevalence.by.age (merged Step 2 simset)



r1.lower.middle: population.by.age (merged Step 2 simset)



r1.lower.middle: cascade (merged Step 2 simset)

awareness

engagement

suppression

value

year

1.00
0.75
0.50
0.25
0.00

1980
2000
2020
2040

sim.id
1
truth

sim.id
1

split
all

r1.lower.middle: cascade.by.age.sex (merged Step 2 simset)

